EXERCISE 8

General Bernoulli Equation Reduction

Original statement · ProblemsCh2.pdf, p. 2

Bernoulli • Substitution z = y^(1-α)

PROBLEM STATEMENT

Show that the Bernoulli equation

$$y'(x) = a(x)\,y(x) + b(x)\,y(x)^{\alpha},$$

where $\alpha \neq 0, 1$, can be transformed into a linear ODE with the change of variable $z(x) = y(x)^{1-\alpha}$ and find the solution. What happens if $\alpha = 0$ or $\alpha = 1$?

Source: ProblemsCh2.pdf, Exercise 8 (page 2).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Assume a and b are continuous on an interval I and $\alpha \in \mathbb{R}$, $\alpha \neq 0, 1$. The power y^{α} and $y^{1-\alpha}$ must make sense, so we work with $y > 0$ (for rational α with odd denominators other branches can be treated analogously). On $y > 0$, $f(x, y) = ay + by^{\alpha}$ and $\partial f/\partial y = a + \alpha b\, y^{\alpha-1}$ are continuous, so IVPs with $y(x_0) = y_0 > 0$ have unique local solutions.

The equation is non-linear because y^{α} with $\alpha \neq 0, 1$ is not affine in y. When $\alpha > 0$, $y \equiv 0$ is also a solution; it is excluded by the assumption $y > 0$ and has to be added by hand (for $0 < \alpha < 1$ uniqueness at $y = 0$ may fail, as in Problem 2.14).

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this substitution: dividing by y^α gives $y^{-\alpha}y' = a\,y^{1-\alpha} + b$, where the left side is, up to a constant factor, the derivative of $y^{1-\alpha}$ and the right side contains $y^{1-\alpha}$ linearly. So the natural unknown is $z = y^{1-\alpha}$.

Change of variables written explicitly: $z = y^{1-\alpha},\ z' = (1-\alpha)\,y^{-\alpha}\,y'$ (chain rule), $y = z^{1/(1-\alpha)}$.

CLASS NOTES AND COMPLEMENT [NOTES, P. 3]

The Bernoulli substitution $z = y^{1-\alpha}$ is a complement (not in the class notes). The linear equation for z is then solved in Step 2 below with the three-step algorithm of the notes (μ there is the notes' I): Step 1 takes $P(x) = \int_{x_0}^{x} p(s)\,ds = -(1-\alpha)A(x)$, so $I(x_0) = 1$; Step 2 integrates $(Iz)' = Iq$ from x_0 to x; Step 3 gives $z = \frac{z_0}{I} + \frac{1}{I}\int_{x_0}^{x} Iq\,ds$, which is the formula of Step 2.

PHASE 3: STEP-BY-STEP DERIVATION

Step 1: derive the linear equation

Differentiate $z = y^{1-\alpha}$ with the chain rule and substitute the equation for y':

$$z' = (1-\alpha)\,y^{-\alpha}\,y' = (1-\alpha)\,y^{-\alpha}\,(a\,y + b\,y^{\alpha}) = (1-\alpha)\left(a\,y^{1-\alpha} + b\right).$$

Since $y^{1-\alpha} = z$:

$$z' - (1-\alpha)\,a(x)\,z = (1-\alpha)\,b(x).$$

This is linear first order in z, with $p(x) = -(1-\alpha)a(x)$ and $q(x) = (1-\alpha)b(x)$, both continuous.

Step 2: solve it with an integrating factor

Fix $x_0 \in I$ and define $A(x) = \int_{x_0}^{x} a(s)\,ds$. Then $\int p\,dx = -(1-\alpha)A(x)$ and

$$\mu(x) = e^{-(1-\alpha)A(x)}, \qquad \mu(x_0) = 1, \qquad (\mu z)' = (1-\alpha)\,b(x)\,\mu(x).$$

Integrate from x_0 to x (Barrow's rule), with $z_0 = z(x_0) = y_0^{1-\alpha}$:

$$\mu(x)z(x) - z_0 = (1-\alpha)\int_{x_0}^{x} b(s)\, e^{-(1-\alpha)A(s)}\, ds.$$

Multiply by $1/\mu(x) = e^{(1-\alpha)A(x)}$:

$$z(x) = e^{(1-\alpha)A(x)}\left[z_0 + (1-\alpha)\int_{x_0}^{x} b(s)\, e^{-(1-\alpha)A(s)}\, ds\right].$$

Step 3: return to y

Since $y = z^{1/(1-\alpha)}$, wherever the bracket is positive (so that $y > 0$):

$$y(x) = \left\{e^{(1-\alpha)A(x)}\left[y_0^{1-\alpha} + (1-\alpha)\int_{x_0}^{x} b(s)\, e^{-(1-\alpha)A(s)}\, ds\right]\right\}^{\frac{1}{1-\alpha}}.$$

The solution exists as long as the expression in braces stays positive; if it reaches 0 at a finite point the solution ceases to exist there (the substitution is valid only for $y > 0$).

Step 4: the excluded cases $\alpha = 0$ and $\alpha = 1$

- $\alpha = 1$: the equation is $y' = (a+b)\,y$, already linear and homogeneous. The substitution would give $z = y^0 = 1$, a constant that carries no information (the formula above has the factor $1-\alpha = 0$, so it degenerates to $z' = 0$). The solution is obtained directly: $y(x) = y_0 \exp\left(\int_{x_0}^{x}(a+b)\, ds\right)$.
- $\alpha = 0$: the equation is $y' = a\,y + b$, already linear (inhomogeneous). The substitution gives $z = y^1 = y$, the identity, and the formula above with $1-\alpha = 1$ reproduces the variation-of-constants solution $y = e^{A}\left[y_0 + \int_{x_0}^{x} b\, e^{-A}\, ds\right]$. No transformation is needed.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Check the general solution by differentiation. Put $B(x) = z_0 + (1-\alpha)\int_{x_0}^{x} b\, e^{-(1-\alpha)A}\, ds$, so $z = e^{(1-\alpha)A}B$ and

$$z' = (1-\alpha)a\, e^{(1-\alpha)A}B + e^{(1-\alpha)A}(1-\alpha)\, b\, e^{-(1-\alpha)A} = (1-\alpha)a\, z + (1-\alpha)b.$$

This is the linear equation of Step 1, and $z(x_0) = e^0 z_0 = z_0$. Undoing the change of variables is reversible for $y > 0$, so $y = z^{1/(1-\alpha)}$ solves the Bernoulli equation.

Consistency with Problem 2.7: for $a=1, b=x, \alpha=-1$ the formula gives $1-\alpha=2$, $A=x$, $z'-2z=2x$, which is the equation obtained there for $z=y^2$.

Check with a constant-coefficient case (a, b constants, $\alpha=2$, so $1-\alpha=-1$): $z'+az=-b$, giving $z=-\frac{b}{a}+\left(z_0+\frac{b}{a}\right)e^{-a(x-x_0)}$ and $y=1/z$, which is the logistic-type solution of $y'=ay+by^2$. A sympy substitution of this closed form into $y'=ay+by^2$ (with $a, b>0$ symbols and $y_0=1$) gives residual 0.

RESULT

$z=y^{1-\alpha}$ turns the Bernoulli equation into $z'-(1-\alpha)az=(1-\alpha)b$, with solution $z=e^{(1-\alpha)A}\left[z_0+(1-\alpha)\int b\,e^{-(1-\alpha)A}\right]$ and $y=z^{1/(1-\alpha)}$. For $\alpha=1$ and $\alpha=0$ the equation is already linear and the substitution is trivial or degenerate.